

CT-268 PROFESSIONAL ETHICS

LECTURE

PROFESSIONAL
RESPONSIBILITIES, RISK,
AND SAFETY

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OCCUPATION vs PROFESSION

- Occupation is an activity undertaken by a person to earn his/her livelihood.
- *Professional* practice requires acquisition of the *special knowledge and skill* peculiar to one's profession and application of that knowledge to achieve certain ends.
- The further requirement for an occupation to be a profession, namely, that the ends it seeks are to preserve or promote some aspect of human well-being.

PROFESSION AND PROFESSIONALISM



- **Professions** are those occupations that both, require advanced study and mastery of a specialized body of knowledge, and undertake to promote, ensure, or safeguard some aspect of others' well-being.
- **Professionalism** is the conduct, aims, or qualities that characterize or mark a profession or a professional person.



CHARACTERISTICS OF PROFESSION

- **Extensive training**
- **Vital knowledge and skills**
- **Control of services**
 - Only graduates of accredited professional institutions should hold the professional title. The profession controls accreditation standards, curriculum, and the number of institutions.
 - A license is required to legally practice the profession. Practicing without a license can result in legal penalties.
- **Autonomy in the workplace**
 - Professionals use individual judgment and creativity in their work, especially in private practice. Based on their specialized knowledge that enables them to choose appropriate services, professionals decide the best approach for their clients or patients, such as medical treatment or legal defense.
 - Another aspect to autonomy can be financial interest; such as when a doctor orders unnecessary tests from a company they have a financial interest in.
- **Claim to ethical regulation**



Identify two professions other than yours. What characteristics do they share such that those occupations are all considered professions?



PROFESSIONAL CONDUCT

- Organization
 - Level-headedness
 - Thoroughness
 - Dedication
 - Attentiveness to details
 - Promptness
 - Taking pride in their profession
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- Clients expect these traits of the person they contract for their project.



YOUR PROFESSION....

- is a group activity, which openly professes special knowledge, skill, and judgment.
- is the occupation by which most of you earn your living.
- is entered into voluntarily.
- serves a morally good end, namely the production of technology for the benefit of mankind.
- has special obligations, including protecting the health and safety of the public, as this is affected by technology.



TWO MODELS OF PROFESSIONALISM

- Another way to understand the importance of the ethical element in professionalism is to examine two models of the professional.
 - The Business Model
 - According to the business model, an occupation is primarily oriented toward making a profit within the boundaries set by law.
 - Professionals use their specialized knowledge and expertise to earn profit and justify higher income or preferential treatment. The main difference is that unlike ordinary businesses that sell physical products, professionals sell their expertise and services for profit.
 - The Professional Model
 - This model offers a quite a different picture of occupations such as medicine, law, and engineering.
 - Professionals are trusted to prioritize public safety, health, and welfare by maintaining high standards of competence and ethics.
 - In return for protecting the public, professionals receive social status, higher income, workplace autonomy, and support for their education.



RESPONSIBILITY

- The state or fact of being accountable for something within one's power, control, or management.
- Reliability, trustworthiness.
- The concept of responsibility is many-faceted.
- As a notion of accountability, it may be applied to individuals, teams, divisions or units within organizations, or even organizations themselves.
 - It may focus primarily on legal liabilities, job-defined roles, or moral accountability.
- Our focus is mainly on the moral accountability of individuals.



- Professionals are expected to commit themselves to high standards of conduct.
- We would expect integrity, honesty, civic-mindedness, and a willingness to make some self-sacrifice.
- In addition to displaying basic competence, a highly responsible professional would be expected to:
 - exhibit imaginativeness and perseverance
 - communicate clearly and informatively
 - be committed to objectivity
 - be open to acknowledging and correcting mistakes
 - work well with others
 - be committed to quality
 - be able to see the “big picture” as well as more minute details
- What all these characteristics have in common is that they contribute to the reliability and trustworthiness of professionals.



STANDARD OF CARE

- Professionals have an obligation to conform to the standard operating procedures and regulations that apply to their profession and to fulfill the basic responsibilities of their job as defined by the terms of their employment.
- Unexpected problems can arise that standard operating procedures and current regulations are not well equipped to handle.
- Professionals who have responsible charge for a project are expected to exercise careful oversight before putting their official stamp of approval on the project.



The Kansas City Hyatt Regency during the First Day of Investigation. The second and fourth floor walkways lie on the lobby floor. They were moved from their original positions to extricate people from the wreckage. The third floor walkway is still suspended at left. (Photo by Lee L. Lowery).



BLAME RESPONSIBILITY AND CAUSATION

- The concept of cause is related in an interesting way to that of responsibility.
- The more we tend to speak of the physical cause of something, the less we tend to speak of responsibility—and the more we tend to speak of responsibility, the less we tend to focus on physical causes.
- If organizations can be causes, can they also be morally responsible agents, much as humans can be?
- Regardless of whether organizations are seen as moral agents or merely causes of harm, organizations can be held responsible in at least three senses.
 - First, they can be criticized for harms.
 - Second, an organization that harms others can be asked to make reparations for wrong done.
 - Finally, an organization that has harmed others is in need of reforms.



PROFESSIONAL RESPONSIBILITY

- To understand the ethics of one's profession one must understand not only what acts are ethically required, permitted, or forbidden but also what is entrusted to one as a member of one's profession, that is, what good outcomes one is expected to marshal one's professional knowledge to achieve.
- The pursuit of those good outcomes is a **moral responsibility**, specifically, a **professional responsibility**. Fulfilling that responsibility may require considering many potentially competing factors.
- The critical thinking and moral reflection you engage in now will provide you with practice and some background for recognizing and fulfilling your professional responsibilities and maintaining your moral integrity.



- As philosopher John Ladd and sociologist Bernard Barber have both argued, **responsible professional practice** combines two elements:
 - *proficiency in the knowledge and skills* of the profession in question, and
 - *concern* for the well-being of others.
- Responsible practice is practice that is conducive to producing desirable *results* in the area entrusted to the profession in question.
- **Official responsibility** is a responsibility that someone is charged to carry out as part of his/her assigned duties.
- Moral responsibility does not reduce to official responsibility.
 - “I was just doing my job” or “I was just doing what I was told” is not a generally valid excuse for unethical behavior of an adult.



IMPEDIMENTS TO RESPONSIBLE ACTION

- **Self-Interest:** Our own interest overshadows professional responsibility and interests of others.
- **Self-Deception:** An intentional avoidance of truths we would find it painful to confront self-consciously.
 - “I’m not really doing this just for myself.” “Everyone takes shortcuts once in a while—it’s the only way one can survive.”
- **Fear:** Fear of acknowledging our mistakes, of losing our jobs, or of some sort of punishment or other bad consequences.
- **Ignorance:** Lack of awareness and deliberate avoidance of vital information.



- **Egocentric Tendencies:** Interpreting situations from very limited perspectives, and failing to realize that others' perspectives can be different from ours in important ways.
- **Microscopic Vision:** Also has limited perspective, but unlike egocentric thinking which tends to be inaccurate, microscopic vision may be highly accurate and precise but our field of vision is greatly limited.
- **Uncritical Acceptance of Authority:** Deference does not really relieve one from at least partial responsibility for those consequences.
- **Groupthink:** Although this may contribute to better decisions (“two heads are better than one”), it also creates tendencies to engage in situations in which groups come to agreement at the expense of critical thinking.

IT IS UNETHICAL, BUT
FORTUNATELY FROM A
BUSINESS ETHICS PERSPECTIVE
IT IS ETHICAL!



oKURF.

“BUGS,” GLITCHES, AND ERRORS AS CENTRAL CONCERNS IN SOFTWARE ENGINEERING



- The central problem for software engineers (and others who design and test software) is that of creating bug-free software.
 - Sometimes the bugs clearly threaten human health and safety.
- **Therac-25 Radiation Overdoses (1985–1987)**
 - A radiation therapy machine delivered lethal overdoses to six patients due to a race condition in the software.
 - When operators rapidly corrected data entry errors, the software erroneously activated the high-energy electron beam without the necessary scattering shield in place.
 - The manufacturer had reused unverified code from older models, removed physical hardware safety interlocks, and dismissed early hospital complaints, assuming software could not fail.



- **Boeing 737 MAX MCAS Failures (2018–2019)**

- Two commercial flights crashed shortly after takeoff, killing 346 people.
- Automated flight-control software (MCAS) relied on a single Angle-of-Attack sensor. A malfunction caused the software to repeatedly force the plane's nose down.
- Boeing had rushed development to compete with rivals, hid the software's power from pilots to avoid costly simulator training, and pressured internal engineers to overlook safety concerns.

- **General Motors Ignition Switch Cover-Up (2000s–2014)**

- Faulty ignition switches shut off moving engines, disabling power steering, brakes, and airbags, leading to at least 124 deaths.
- The switch could be easily jarred out of the "run" position by a heavy keychain or a bump on the road.
- GM engineers had discovered the defect during development but approved it anyway because it was cheaper. For a decade, the company quietly tracked the deaths while actively denying the defect to families and investigators.



- **Volkswagen "Defeat Device" Scandal (2015)**

- Over 11 million diesel vehicles were sold worldwide with software intentionally designed to cheat emissions tests.
- Software engineers programmed a defeat device that used steering and speed inputs to detect when the car was on a laboratory test stand and temporarily reduce toxic gas output.
- Corporate leadership had ordered the cheating to meet strict US environmental laws without sacrificing car performance. Engineers willingly coded a deliberate deception that resulted in massive environmental pollution and billions in fines.





SAFETY

- Thus far, we have considered mainly those safety hazards that threaten to cause accidents and malfunctions.
- It is not possible to eliminate certain hazards.
 - Some can be eliminated or mitigated only by producing other adverse consequences.
 - For example, an effective ingredient in insect repellents, commonly known as DEET, is known to be toxic to humans.
 - People continue to use repellents containing DEET because they have not found less dangerous substances that are equally effective in repelling insects.
 - Insect bites may carry serious diseases like dengue fever.
 - The use of DEET in some circumstances is justified by health considerations.
 - The decision to use DEET involves a tradeoff between risks and benefits.



- There are least two major considerations governing the evolution of requirements for professionals to fulfill their responsibility for safety.
 - First, professionals' responsibility for safety extends only as far as the possible outcomes that they can foresee at a given point in the development of knowledge.
 - Second, the professionals have a responsibility to examine those results that (s)he might control or influence.
- To be morally responsible for outcomes, people must have some ability to foresee and influence them.
 - Some critics have sought to blame technology (and engineers/computer professionals) for everything that is objectionable in modern life.
 - Certainly technology has had harmful unintended consequences. However, the same is true for policies and decisions in other spheres as well.



RISK

- Every profession involves risk, and innovation usually increases the risks.
 - Without innovation, there is no progress.
- One cannot avoid risk simply by remaining with tried and true designs.
 - But innovation creates technologies in which the risks are not fully understood, which increases the chance of failure.
- The relationship of safety to risk is an inverse one.
- Risk is “a compound measure of the probability and magnitude of adverse effect.”
 - It is composed of two elements:
 - The likelihood of an adverse effect or harm and
 - The magnitude of that adverse effect or harm.
- Because of the laws of science and statistics, the more we accept risk in a project, the less safe it will become.
 - If there were absolutely no risk in a project, then that project would be absolutely safe.



FACTORS OF SAFETY

- One of the most common concepts in professional practice is the notion of “factors of safety.”
- In computing, it translates to building extra capacity and fault tolerance.
 - **Capacity Margins:** Designing servers or databases to handle traffic loads well above peak estimates (e.g., handling 10x the daily traffic for Azadi Sale event).
 - **Redundancy (Fail-safes):** Using backup servers, cloud mirroring, and parallel systems so that if one system breaks, another takes over instantly.
 - **Error Budgets:** Creating an acceptable limit for system downtime or bugs. If a system requires 99.999% uptime, developers build in a tight "error budget" to test new features without risking total system failure.



THE PROFESSIONAL'S APPROACH TO RISK

- **Risk as the Product of the Probability and Magnitude of Harm**
 - Professionals and other experts on risk often believe that the public is confused about risk, because the public does not have the correct factual information about the likelihood of certain harms.
- **Acceptable Risk**
 - An acceptable risk is one in which the product of the probability and magnitude of the harm is equaled or exceeded by the product of the probability and magnitude of the benefit, and there is no other option where the product of the probability and magnitude of the benefit is substantially greater.



- **The Capabilities Approach to Identifying Harm and Benefit**

- There are four main limitations with quantifying harm:
 - First: Hazard impacts include not only immediate losses (e.g., deaths, power outages) but also broader, indirect societal harms.
 - Second: Hazards can create opportunities as well as harms; ignoring benefits may overestimate their overall negative impact.
 - Third: A consistent metric is needed to accurately measure both the harms and benefits of hazards, including nonfatal, psychological, and indirect effects.
 - Fourth: Current methods do not clearly link specific losses to changes in individual or societal well-being and, ultimately, quality of life.



THE PUBLIC'S APPROACH TO RISK

- **Expert and Layperson: Differences in Factual Beliefs**

- The public and risk experts often view risk very differently, causing miscommunication.
- Experts believe people may misjudge the probability of death or injury from certain activities or technologies.
- People tend to overestimate low-probability risks and underestimate high-probability risks.

- **Risky Situations and Acceptable Risk**

- Public often blends risk with acceptable risk, unlike professionals who distinguish them clearly.
- “Risky” is commonly used as a warning for something unfamiliar, untrustworthy, or requiring caution—not just to describe probability.
- People assess risk using factors beyond expected deaths or injuries, such as financial rewards, trust, and familiarity.



- **Free and Informed Consent**

- Free and informed consent requires three conditions: no coercion, relevant information, and the ability to understand and evaluate the risks.
- Meaningful consent is difficult because people may lack real alternatives, adequate information, or accurate risk understanding.
- For risks affecting communities, silence or lack of protest does not necessarily mean consent, especially when people are unaware of the risks.
- Compensation after harm can avoid the need for prior consent, but it is difficult to determine fair compensation, may restrict individual freedom, and cannot adequately compensate for severe injury or death.
- Both consent and compensation have limitations, but efforts should still be made to respect the moral agency of people exposed to technological risks.

- **Equity or Justice**

- The ethics of respect for persons, places great emphasis on respecting the moral agency of individuals, regardless of the cost to the larger society.

THE GOVERNMENT REGULATOR'S APPROACH TO RISK



- Regulators face a dilemma regarding risk management.
- On one hand, regulators could decide to regulate only when there is a provable connection between a substance and some undesirable effect such as a risk of cancer.
 - Because of the difficulties in establishing the acceptable levels of exposure to toxic substances at which there is no danger, this option would expose the public to unacceptable risks.
- On the other hand, regulators could eliminate any possible risk insofar as this is technologically possible.
 - Choosing this option would result in the expenditure of large sums of money to eliminate minute amounts of any substance that might possibly pose risks to human beings.
 - This would not be cost-effective.
 - Funds might better be spent elsewhere to eliminate much greater threats to public health.
- From the standpoint of the government regulator:
 - An acceptable risk is one in which protecting the public from harm has been weighted more heavily than benefiting the public.



THE PROFESSIONAL'S LIABILITY FOR RISK

- There are at least two issues here:
 - One is that the standards of proof in tort law and science are different, and this produces an interesting ethical conflict.
 - Another issue is that in protecting the public from unnecessary risk, engineers may themselves incur legal liabilities.
- **The Standards of Tort Law**
 - Litigation that seeks redress from harm most commonly appeals to the law of torts, which deals with injuries to one person caused by another, usually as a result of fault or negligence of the injuring party.
 - Many of the most famous legal cases involving claims of harm from technology have been brought under the law of torts.
 - The litigation involving harm from asbestos is one example.



IN PAKISTAN...

- While the core areas of tort law in computing *internationally* include negligence, professional malpractice, product liability, and fraud and misrepresentation, it is underdeveloped and rarely litigated in its pure form *in Pakistan*.
- Instead, liability for tech professionals is primarily enforced through statutory laws, contract law, and consumer protection courts.
- Section 9 of the *Code of Civil Procedure (CPC)*, *Prevention of Electronic Crimes Act, 2016 (PECA)*, *Electronic Transactions Ordinance, 2002 (ETO)*, *Consumer Courts*, and Breach of Contract under the *Contract Act, 1872* can be used.
- That is why, software houses in Karachi, Lahore, and Islamabad heavily rely on strict *Service Level Agreements (SLAs)* and *limitation-of-liability* clauses.
 - These contracts explicitly state that the developer is not liable for indirect financial losses caused by system downtime or bugs.

BECOMING A RESPONSIBLE PROFESSIONAL REGARDING RISK



- The development of new technology is intimately connected with risk.
- The obligation of professionals is to be ethically responsible with regard to risk.
- The first step in the process of becoming ethically responsible about risk is to be aware of the fact that risk is often difficult to estimate and can be increased in ways that may be indirect and deceitful.
- The second step is to be aware that there are different approaches to the determination of acceptable risk.
- The third step is to assume their responsibility, as the experts in technology, to communicate issues regarding risk to the public, with the full awareness that both the public and government regulators have a somewhat different agenda with regard to risk.



- Given these considerations, a more general principle of acceptable risk may be constructed, which may provide some guidance in determining when a risk is within the bounds of moral permissibility:
 - *People should be protected from the harmful effects of technology, especially when the harms are not consented to, or when they are unjustly distributed, except that this protection must sometimes be balanced against:*
 - *The need to preserve great and irreplaceable benefits, and*
 - *The limitation on our ability to obtain informed consent.*

